Quantum BCH $[[15, 7, 3]]$ Code

1. Introduction

The $[[15, 7, 3]]$ Quantum Bose-Chaudhuri-Hocquenghem (BCH) code is a foundational example of a high-rate quantum error-correcting code. It belongs to the family of cyclic CSS codes and is constructed using the classical $[15, 11, 3]$ Hamming code.

The most notable feature of this code is its **Code Rate (k/n)**. While the Steane code encodes 1 logical qubit into 7 physical qubits (rate ≈ 0.14), the $[[15, 7, 3]]$ BCH code encodes **7 logical qubits** into 15 physical qubits (rate ≈ 0.47). This high density makes algebraic block codes highly desirable for resource-efficient fault-tolerant architectures.

1.1 The Parity Check Matrix

The stabilizer generators are defined by the parity check matrix of the classical $[15, 11, 3]$ code. This matrix H is a 4×15 binary matrix where the j-th column is simply the binary representation of the integer j.

$$H = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

Because all rows have an even weight of 8, and the overlap between any two rows is exactly 4, the matrix is self-orthogonal ($HH^T = 0 \pmod 2$). This perfect orthogonality satisfies the CSS condition, allowing us to use this identical 4×15 matrix for both X-stabilizers and Z-stabilizers.

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2. Qiskit Implementation

This implementation builds a 23-qubit system (15 Data + 8 Ancilla). The decoding process leverages the mathematical synergy of the H matrix: **the binary value of the measured syndrome points directly to the index of the faulty qubit.**

In [5]:
```
%pip install numpy==1.26.4 scipy==1.12.0

import numpy as np
import matplotlib.pyplot as plt
from qiskit import QuantumCircuit, QuantumRegister, ClassicalRegister
from qiskit_aer import AerSimulator
from scipy.optimize import fsolve
```

```
print("Imports successful")
```

```
Requirement already satisfied: numpy==1.26.4 in c:\users\user\anaconda3\lib\site-pac
kages (1.26.4)
Requirement already satisfied: scipy==1.12.0 in c:\users\user\anaconda3\lib\site-pac
kages (1.12.0)
Note: you may need to restart the kernel to use updated packages.
Imports successful
```

In [6]:

```
class QuantumBCH15:
    def __init__(self):
        # 15 Data Qubits
        self.data = QuantumRegister(15, name='data')

        # 8 Ancilla Qubits (4 for Z-checks, 4 for X-checks)
        self.anc = QuantumRegister(8, name='anc')

        # Classical Registers for Syndromes
        self.sz = ClassicalRegister(4, name='sz') # Z-syndromes (detect X flips)
        self.sx = ClassicalRegister(4, name='sx') # X-syndromes (detect Z flips)

        # Final readout
        self.out = ClassicalRegister(15, name='readout')

        self.qc = QuantumCircuit(self.data, self.anc, self.sz, self.sx, self.out)

    def encode_logical_zero(self):
        r"""
        Encodes |0>^(\otimes 7) into the logical CSS state.
        We prepare the uniform superposition of the dual code C^\perp.
        The power-of-2 columns (1, 2, 4, 8) form the basis generators.
        """
        # Put the basis generator qubits into |+>
        # Qubit indices (0-based) for cols 1, 2, 4, 8 are 0, 1, 3, 7
        generators = [0, 1, 3, 7]
        self.qc.h([self.data[q] for q in generators])

        # Entangle the remaining qubits based on their binary column index
        # For any column C, we CNOT from the generators that sum to C
        for col in range(1, 16):
            q_idx = col - 1
            if q_idx in generators:
                continue

            # Binary breakdown of the column index
            if col & 1: self.qc.cx(self.data[0], self.data[q_idx]) # 1
            if col & 2: self.qc.cx(self.data[1], self.data[q_idx]) # 2
            if col & 4: self.qc.cx(self.data[3], self.data[q_idx]) # 4
            if col & 8: self.qc.cx(self.data[7], self.data[q_idx]) # 8

        self.qc.barrier(label='Encoded')

    def measure_syndromes(self):
        """
        Non-destructive measurement of the 8 stabilizers.
```

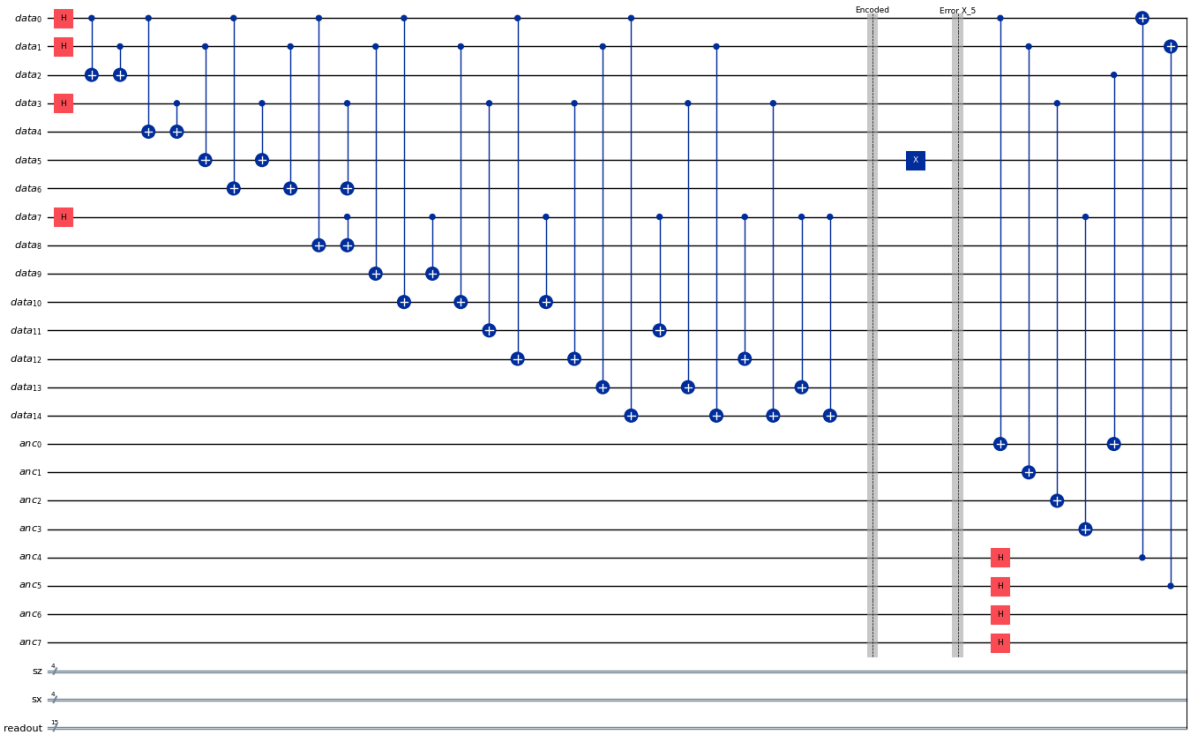
```python
        """
        # Z-Checks (Detect X errors)
        # Row 1 (LSB): Alternating bits (cols 1, 3, 5...)
        for q in [0, 2, 4, 6, 8, 10, 12, 14]: self.qc.cx(self.data[q], self.anc[0])
        # Row 2: (cols 2, 3, 6, 7...)
        for q in [1, 2, 5, 6, 9, 10, 13, 14]: self.qc.cx(self.data[q], self.anc[1])
        # Row 3: (cols 4, 5, 6, 7...)
        for q in [3, 4, 5, 6, 11, 12, 13, 14]: self.qc.cx(self.data[q], self.anc[2]
        # Row 4 (MSB): (cols 8, 9, 10...)
        for q in [7, 8, 9, 10, 11, 12, 13, 14]: self.qc.cx(self.data[q], self.anc[3

        # X-Checks (Detect Z errors)
        self.qc.h(self.anc[4:8])

        for q in [0, 2, 4, 6, 8, 10, 12, 14]: self.qc.cx(self.anc[4], self.data[q])
        for q in [1, 2, 5, 6, 9, 10, 13, 14]: self.qc.cx(self.anc[5], self.data[q])
        for q in [3, 4, 5, 6, 11, 12, 13, 14]: self.qc.cx(self.anc[6], self.data[q]
        for q in [7, 8, 9, 10, 11, 12, 13, 14]: self.qc.cx(self.anc[7], self.data[q

        self.qc.h(self.anc[4:8])

        # Measurements
        self.qc.measure(self.anc[0:4], self.sz)
        self.qc.measure(self.anc[4:8], self.sx)
        self.qc.barrier(label='Syndrome Extracted')

    def apply_correction(self):
        """
        Applies dynamic correction. The syndrome directly matches the 1-based
        column index
        """
        # Correct X Errors using Z-syndrome (Values 1 to 15)
        for i in range(1, 16):
            with self.qc.if_test((self.sz, i)):
                self.qc.x(self.data[i-1])

        # Correct Z Errors using X-syndrome (Values 1 to 15)
        for i in range(1, 16):
            with self.qc.if_test((self.sx, i)):
                self.qc.z(self.data[i-1])

        self.qc.barrier(label='Corrected')

    def decode_and_verify(self):
        r"""
        Uncomputes the encoding sequence. If correct, returns to |0>^{times 15}
        """
        generators = [0, 1, 3, 7]

        # Uncompute in reverse (though CNOTs commute here)
        for col in range(15, 0, -1):
            q_idx = col - 1
            if q_idx in generators:
                continue

            if col & 8: self.qc.cx(self.data[7], self.data[q_idx])
```

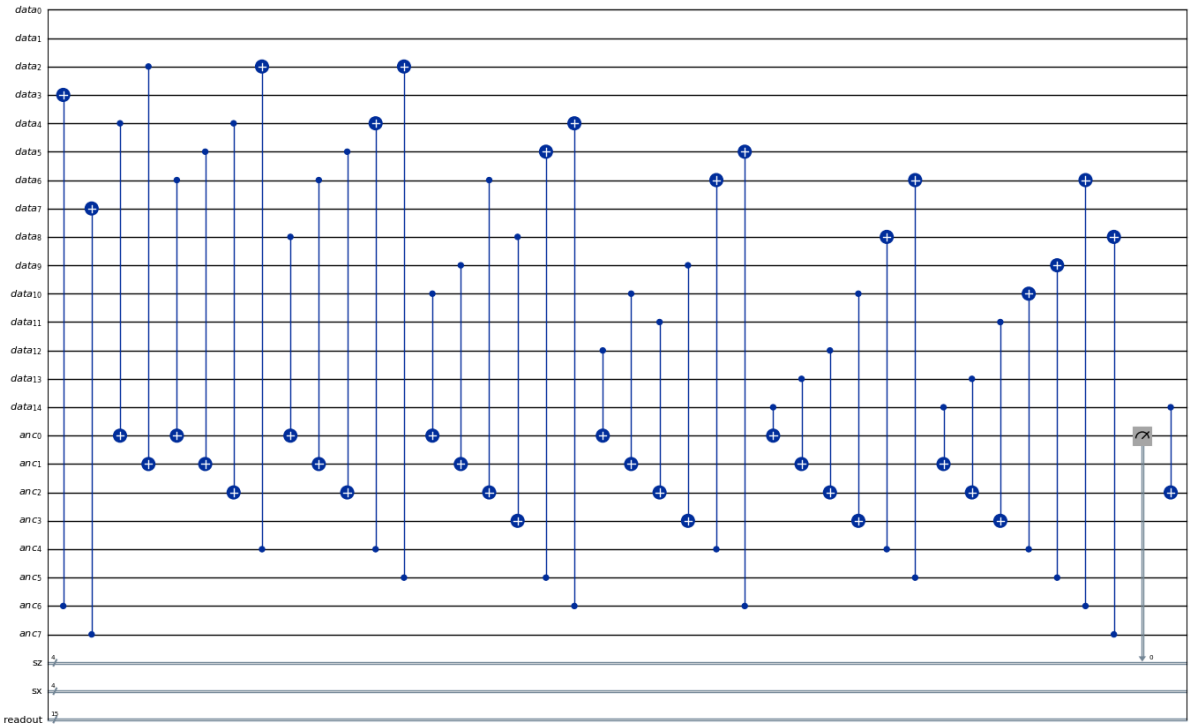
```
            if col & 4: self.qc.cx(self.data[3], self.data[q_idx])
            if col & 2: self.qc.cx(self.data[1], self.data[q_idx])
            if col & 1: self.qc.cx(self.data[0], self.data[q_idx])

        self.qc.h([self.data[q] for q in generators])
        self.qc.measure(self.data, self.out)

        return self.qc
```

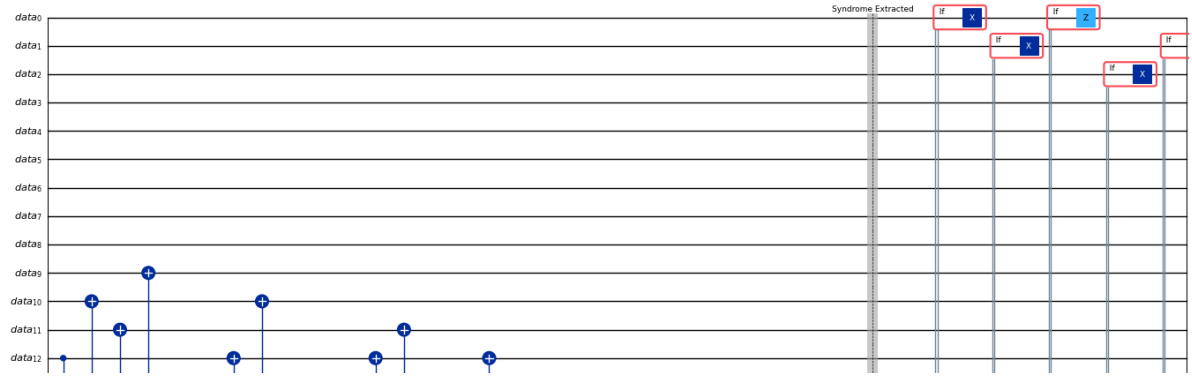
In [7]:

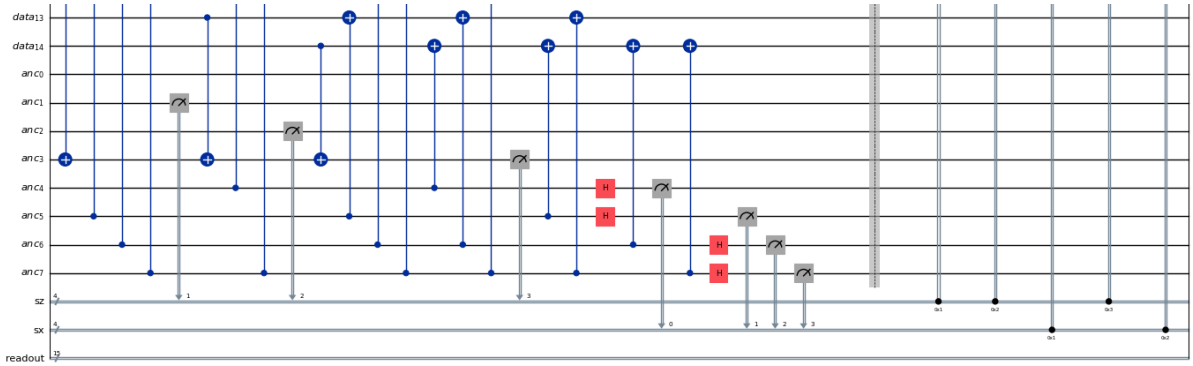
```
bch_demo = QuantumBCH15()
bch_demo.encode_logical_zero()

# Inject a manual error for demonstration (e.g., X error on Qubit 5)
print("Injecting X error on Qubit 5...")
bch_demo.qc.x(bch_demo.data[5])
bch_demo.qc.barrier(label='Error X_5')

# Complete the cycle
bch_demo.measure_syndromes()
bch_demo.apply_correction()
bch_circ = bch_demo.decode_and_verify()

# Draw the circuit
# scale=0.5 is necessary here for fitting screen
display(bch_circ.draw('mpl', fold=40, scale=0.5))
```

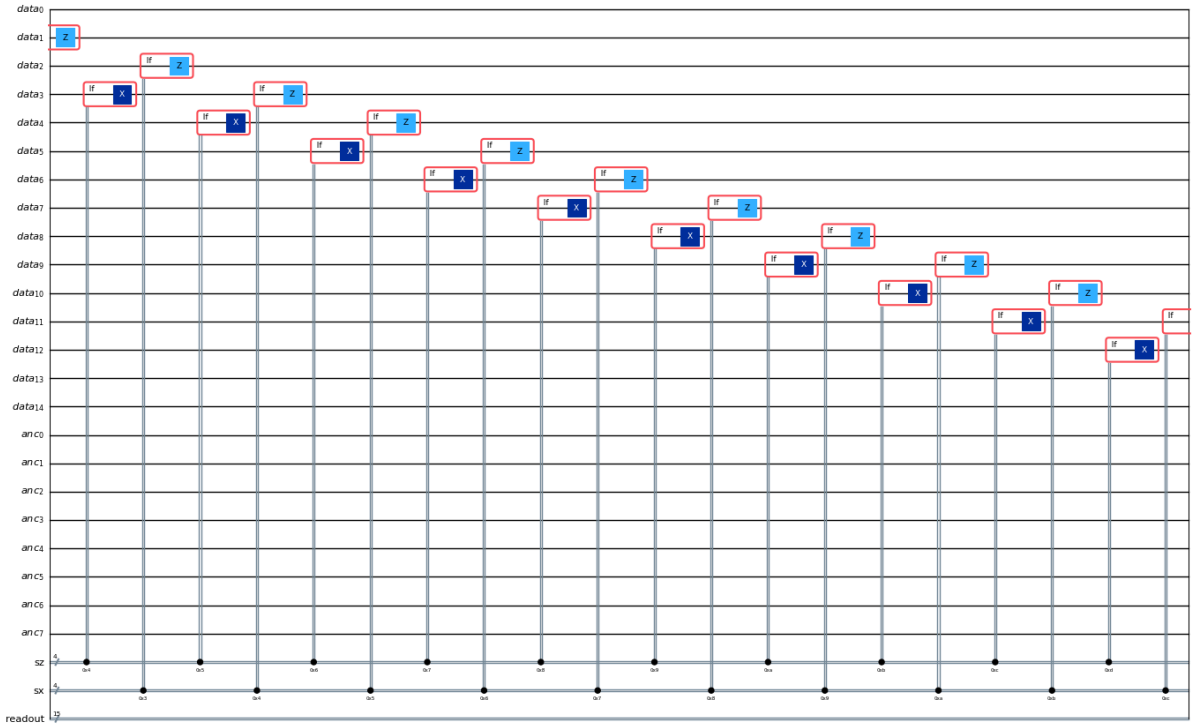
```
Injecting X error on Qubit 5...
```

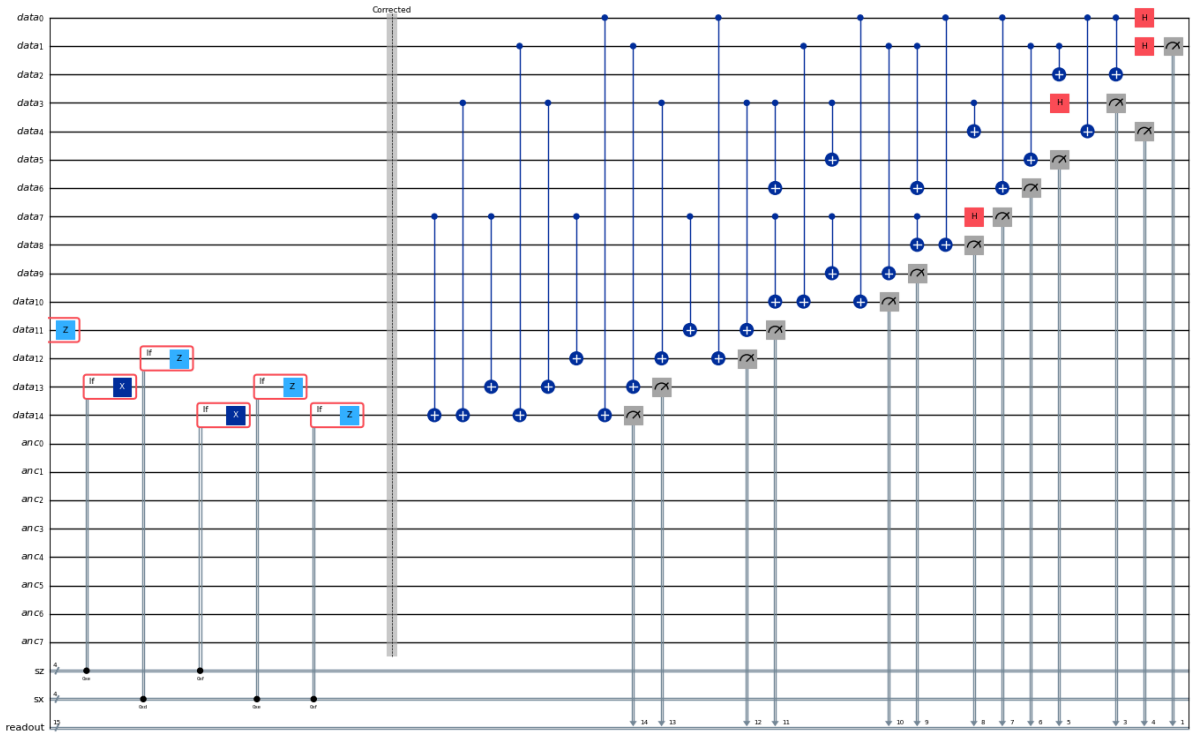

Encoded
Error X_5
data0
data1
data2
data3
data4
data5
data6
data7
data8
data9
data10
data11
data12
data13
data14
anc0
anc1
anc2
anc3
anc4
anc5
anc6
anc7
sz
sx
readout


data0
data1
data2
data3
data4
data5
data6
data7
data8
data9
data10
data11
data12
data13
data14
anc0
anc1
anc2
anc3
anc4
anc5
anc6
anc7
sz
sx
readout


Syndrome Extracted
data0
data1
data2
data3
data4
data5
data6
data7
data8
data9
data10
data11
data12


data13
data14
anc0
anc1
anc2
anc3
anc4
anc5
anc6
anc7
sz
sx
readout


data0
data1
data2
data3
data4
data5
data6
data7
data8
data9
data10
data11
data12
data13
data14
anc0
anc1
anc2
anc3
anc4
anc5
anc6
anc7
sz
sx
readout
If
X
Z


Corrected
data0
data1
data2
data3
data4
data5
data6
data7
data8
data9
data10
data11
data12
data13
data14
anc0
anc1
anc2
anc3
anc4
anc5
anc6
anc7
sz
sx
readout
If
X
Z
H

---

3. Performance Simulation

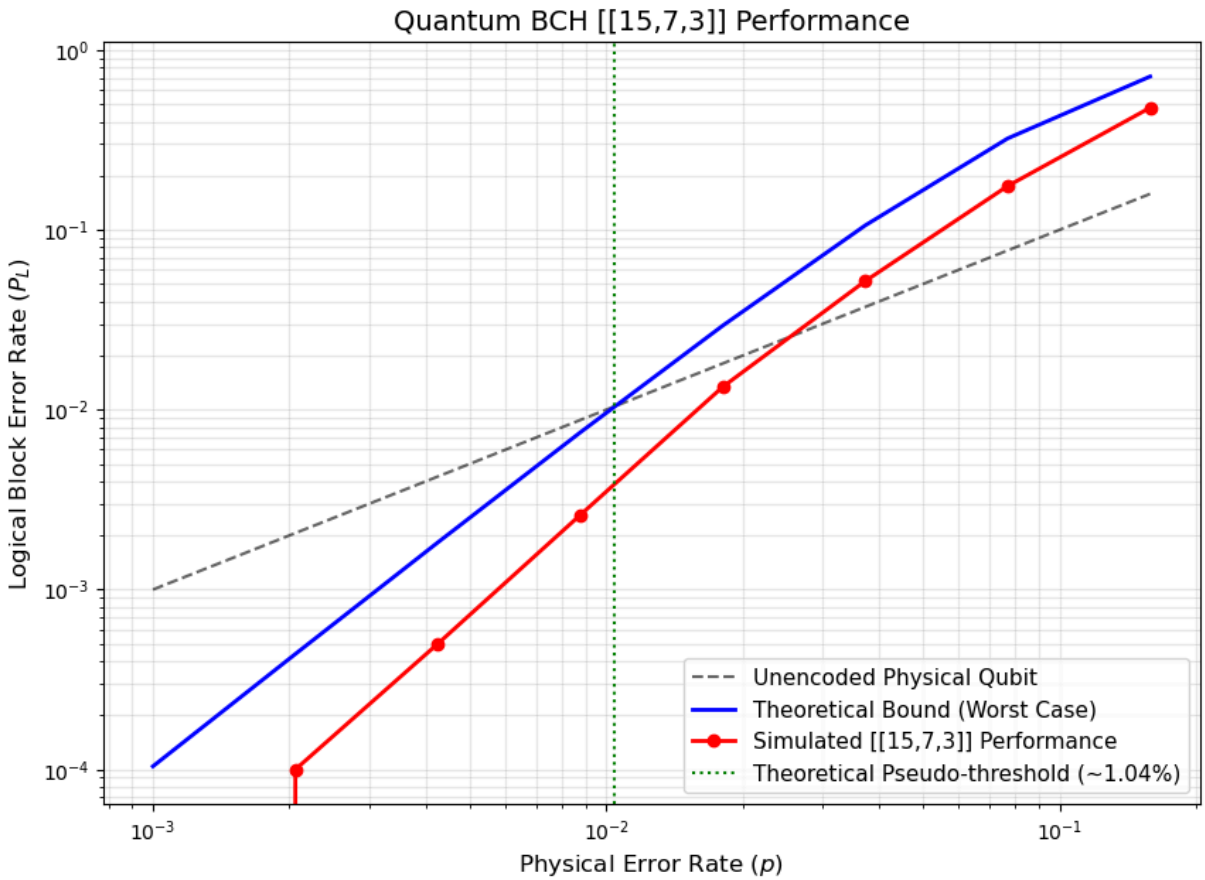
To evaluate the $[[15, 7, 3]]$ code, we inject independent depolarizing noise (X, Y, or Z) on all 15 data qubits.

Because it encodes 7 logical qubits simultaneously, the definition of a "logical failure" is strict: if **any** of the 7 logical qubits is corrupted, the block fails. The theoretical worst-case failure probability for a 15-qubit block correcting 1 error is:

$$P_{theory} = 1 - (1-p)^{15} - 15p(1-p)^{14}$$

In [8]:
```
def apply_random_noise(bch_instance, p):
    """Applies depolarizing noise across all 15 physical qubits."""
    for qubit_idx in range(15):
        if np.random.random() < p:
            error_type = np.random.choice(['X', 'Y', 'Z'])
            if error_type == 'X': bch_instance.qc.x(bch_instance.data[qubit_idx])
            elif error_type == 'Y': bch_instance.qc.y(bch_instance.data[qubit_idx])
            elif error_type == 'Z': bch_instance.qc.z(bch_instance.data[qubit_idx])

def run_bch_simulation(p_values, shots_per_point=1000):
    logical_error_rates = []
    backend = AerSimulator(method='stabilizer')

    print(f"Simulating {len(p_values)} points for [[15,7,3]] BCH Code...")
    for p in p_values:
        failures = 0
```

```
        circuits = []

        for _ in range(shots_per_point):
            bch = QuantumBCH15()
            bch.encode_logical_zero()
            apply_random_noise(bch, p)
            bch.measure_syndromes()
            bch.apply_correction()
            circuits.append(bch.decode_and_verify())

        result = backend.run(circuits, shots=1, optimization_level=0).result()

        for i in range(shots_per_point):
            counts = result.get_counts(i)
            outcome = list(counts.keys())[0]
            readout = outcome.split(" ")[0]

            # Success means all 15 qubits returned to 0
            if readout != '000000000000000':
                failures += 1

        p_logical = failures / shots_per_point
        logical_error_rates.append(p_logical)
        print(f"p={p:.4f} -> P_Logical={p_logical:.4f}")

    return logical_error_rates

# Define physical error rates
p_phys_samples = np.logspace(-3, -0.8, 8)

# Run simulation
p_log_simulated = run_bch_simulation(p_phys_samples, shots_per_point=10000)

# Calculate Theoretical Bounds
# Worst-case failure: 2 or more errors anywhere in the 15-qubit block
p_theoretical = 1 - (1 - p_phys_samples)**15 - 15 * p_phys_samples * (1 - p_phys_sa

# Plotting
plt.figure(figsize=(10, 7))

plt.loglog(p_phys_samples, p_phys_samples, 'k--', label='Unencoded Physical Qubit',
plt.loglog(p_phys_samples, p_theoretical, 'b-', label='Theoretical Bound (Worst Cas
plt.loglog(p_phys_samples, p_log_simulated, 'ro-', label='Simulated [[15,7,3]] Perf

plt.title('Quantum BCH [[15,7,3]] Performance', fontsize=14)
plt.xlabel('Physical Error Rate ($p$)', fontsize=12)
plt.ylabel('Logical Block Error Rate ($P_L$)', fontsize=12)
plt.grid(True, which="both", ls="-", alpha=0.3)

# Calculate the exact crossover point for n=15
def bch_theoretical_diff(p):
    return (1 - (1 - p)**15 - 15 * p * (1 - p)**14) - p

exact_bch_crossover = fsolve(bch_theoretical_diff, 0.01)[0]

plt.axvline(x=exact_bch_crossover, color='green', linestyle=':',
```

```
            label=f'Theoretical Pseudo-threshold (~{exact_bch_crossover*100:.2f}%)'

plt.legend(fontsize=11)

plt.show()
```

```
Simulating 8 points for [[15,7,3]] BCH Code...
p=0.0010 -> P_Logical=0.0000
p=0.0021 -> P_Logical=0.0001
p=0.0043 -> P_Logical=0.0005
p=0.0088 -> P_Logical=0.0026
p=0.0181 -> P_Logical=0.0134
p=0.0373 -> P_Logical=0.0519
p=0.0769 -> P_Logical=0.1753
p=0.1585 -> P_Logical=0.4777
```